Diagnostic Test Accuracy

2026-04-17

Introduction

When a new diagnostic test is proposed, the first question is: how accurately does it classify disease status against a reference standard? The answer comes in several complementary numbers – sensitivity, specificity, positive and negative predictive values, likelihood ratios, and the area under the ROC curve. Each answers a different question, and reporting any one alone is inadequate.

Prerequisites

The reader should understand the distinction between a test result (positive/negative) and disease status (truly positive/negative), and should be comfortable with 2x2 tables and proportions.

Theory

Given a binary test and a binary gold-standard diagnosis, every study produces a 2x2 table:

	Disease +	Disease -
Test +	TP	FP
Test -	FN	TN

From this table:

- **Sensitivity** = TP / (TP + FN). The probability that the test is positive in a diseased person.
- **Specificity** = TN / (TN + FP). The probability that the test is negative in a healthy person.
- **Positive predictive value (PPV)** = TP / (TP + FP). The probability of disease given a positive test.
- **Negative predictive value (NPV)** = TN / (TN + FN). The probability of no disease given a negative test.
- **Positive likelihood ratio (LR+)** = sensitivity / (1 - specificity). How many times more likely a positive test is in diseased versus healthy people.
- **Negative likelihood ratio (LR-)** = (1 - sensitivity) / specificity.

Sensitivity and specificity are properties of the test that are (approximately) invariant to prevalence. Predictive values depend strongly on prevalence: a test with 99% sensitivity and 99% specificity still has PPV below 50% when disease prevalence is 1%. Likelihood ratios, via Bayes' theorem, convert a pre-test probability into a post-test probability and thus tie the test-level quantities to the clinical reasoning a doctor actually does.

For a continuous marker, each possible threshold produces a pair (sensitivity, 1 - specificity). Plotting these across all thresholds gives the **receiver operating characteristic (ROC) curve**. The **area under the ROC curve (AUC)** is the probability that a randomly chosen diseased person has a higher marker value than a randomly chosen healthy person – an interpretable summary of discrimination independent of any threshold.

Assumptions

- The reference standard is a true gold standard (otherwise sensitivity and specificity are biased).

- The test is evaluated on a representative spectrum of diseased and healthy individuals (spectrum bias can inflate apparent accuracy).
- Test results are read blinded to the reference standard (verification and review bias are the two most common threats in reporting).

R Implementation

```r
library(pROC)
library(cutpointr)

set.seed(2026)
n <- 200
disease <- rbinom(n, 1, 0.3)
marker <- ifelse(disease == 1,
                 rnorm(n, mean = 60, sd = 12),
                 rnorm(n, mean = 45, sd = 10))

df <- data.frame(disease = factor(disease, levels = c(0, 1),
                                  labels = c("healthy", "diseased")),
                 marker = marker)

roc_obj <- roc(df$disease, df$marker,
               levels = c("healthy", "diseased"), direction = "<")

auc(roc_obj)
ci.auc(roc_obj)

plot(roc_obj, print.auc = TRUE, ci = TRUE)

cp <- cutpointr(df, marker, disease,
                pos_class = "diseased",
                method = maximize_metric,
                metric = youden)
summary(cp)
plot(cp)
```

`pROC::roc()` constructs the ROC object from marker values and disease labels. `auc()` and `ci.auc()` give the point estimate and 95% CI. The `cutpointr` package finds the threshold that maximises Youden's index (sensitivity + specificity - 1) and reports the operating characteristics at that threshold.

Output & Results

The simulated example gives an AUC of approximately 0.85 (95% CI 0.79 to 0.90). The Youden-optimal cutoff is around 52, with sensitivity ~0.75 and specificity ~0.80 at that threshold.

Interpretation

A manuscript table should report sensitivity, specificity, PPV, NPV, LR+, LR-, and the AUC, each with 95% confidence intervals. For a binary test:

"Sensitivity was 75% (95% CI 66-83%), specificity 80% (74-86%), PPV 62% (52-72%), NPV 88% (82-93%), LR+ 3.8 (2.7-5.2), LR- 0.31 (0.22-0.44), AUC 0.85 (0.79-0.90)."

Crucially, the PPV depends on the disease prevalence in the study population. If the intended clinical use is in a lower-prevalence setting, report the projected PPV at that prevalence using Bayes' theorem.

Practical Tips

- Always report the reference standard explicitly and justify it as a gold standard.
- Report sensitivity and specificity *with* predictive values, not instead of them. Predictive values are what the clinician uses; sensitivity and specificity are what the test provides.
- Use 95% Wilson or Clopper-Pearson intervals for proportions, not the Wald interval, which can extend outside $[0, 1]$ or have poor coverage near 0 and 1.
- Avoid choosing a threshold from the same data that will report its performance; hold out a validation set or use cross-validation.
- Follow STARD reporting guidelines: flow diagram, blinding, reference-standard description, thresholds, and indeterminate results.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale